

# Smart Pantry AI

AI-powered pantry management, expiry tracking,  
and recipe recommendations

# Problem & Project Objective

Why Smart Pantry AI was developed

## Food gets forgotten

Users may lose track of what they already have at home.

## Expiry causes waste

Food can expire before users remember to consume it.

## Meal planning is difficult

It can be hard to decide what to cook from available ingredients.

- Objective: create one web application that manages pantry inventory and uses AI to help users make better use of available food.
- The system combines pantry data, expiry information and generative AI.

# Main Features

What the completed application can do

## Secure accounts

Registration, login  
and user-specific pantry  
data

## Pantry CRUD

Add, view, update and  
delete pantry items

## Expiry tracking

Fresh, Expiring Soon  
and Expired status

## Pantry dashboard

Inventory summary  
and pantry overview

## Ask Pantry AI

Context-aware  
recommendations using  
pantry data

## AI recipes

Generate recipes from  
available ingredients

## Food safety

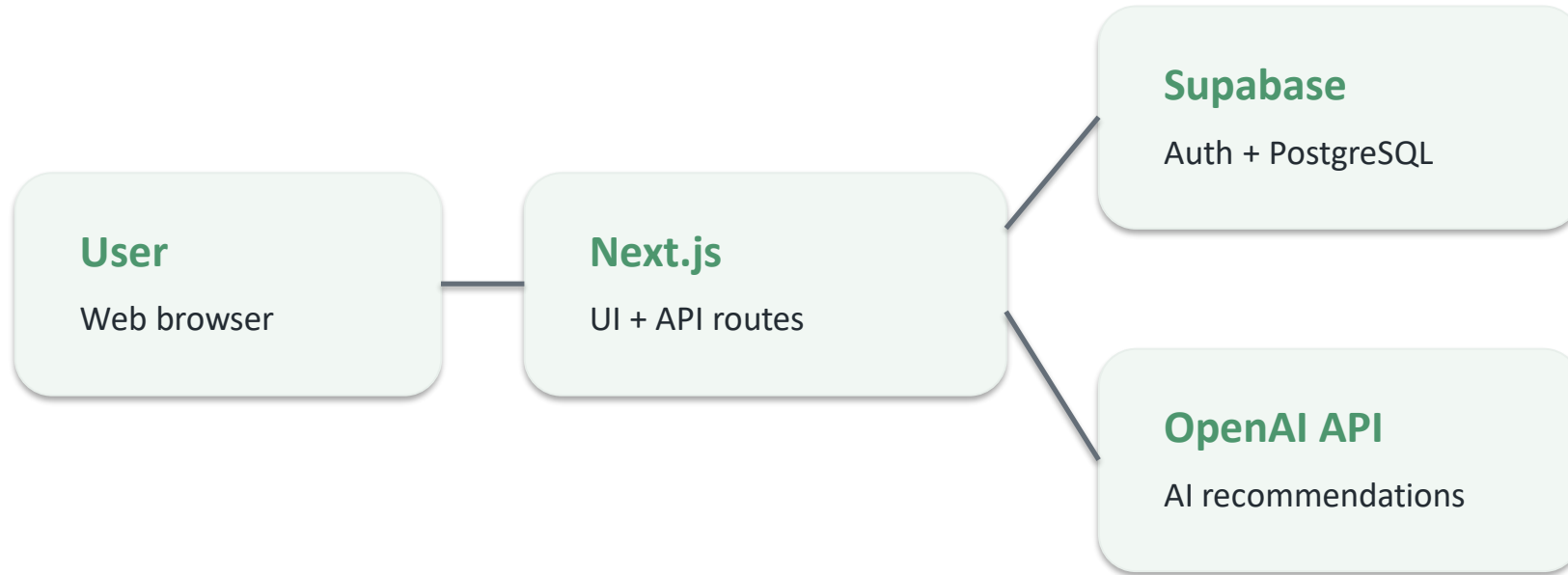
Expired items are excluded  
From consumption  
recommendations

## Cloud-ready

Full-stack web application  
with protected API requests

# Technology & Architecture

Simple full-stack architecture



- Next.js + React + TypeScript
- Supabase Authentication and PostgreSQL with Row Level Security
- OpenAI API for pantry-aware recommendations and recipe generation
- Environment variables protect sensitive credentials

# Testing, Results & Conclusion

What the project demonstrates

## Testing completed

- Authentication
- Pantry CRUD
- Expiry tracking
- AI recommendations
- Recipe generation
- Expired-food safety
- API error handling
- Access control

## Project outcome

The core Smart Pantry workflow was implemented successfully, connecting user pantry data with AI-generated recommendations while maintaining user-specific access controls.

## Future improvements

- Barcode / image-assisted input
- Richer personalization
- Notifications
- More advanced analytics
- Additional AI evaluation

**Smart Pantry AI demonstrates how generative AI can be integrated into a practical full-stack application to reduce food waste and support everyday meal planning.**